Relational Attention for Data-Efficient Language Modeling

Relational BabyLM

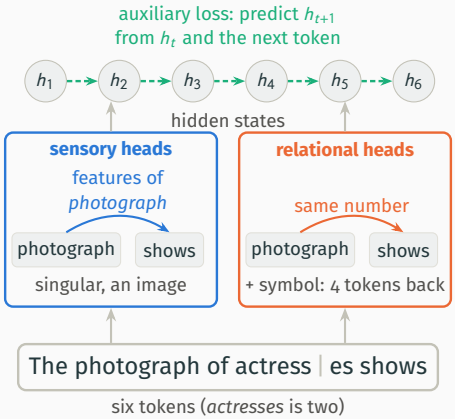
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Relational attention and RNN-like aux loss help a data-limited language model

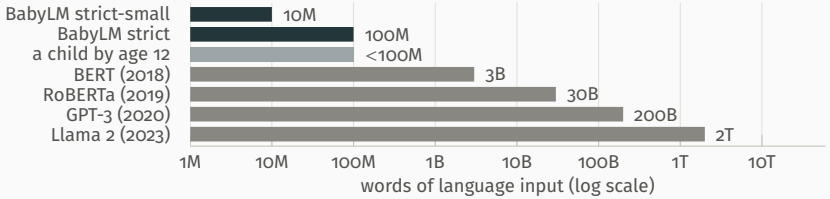
What we build: a language model of the GPT kind, a decoder-only Transformer trained to predict the next token, with two changes:

- some attention heads are relational: they pass on how a token compares with an earlier one, plus a symbol, a tag for the earlier token, in place of its content; the other attention heads are standard self-attention—sensory attn (Webb et al., 2024; Altabaa et al., 2024; Altabaa and Lafferty, 2025);
- an auxiliary loss training each hidden state to work like the state of a recurrent network (RNN): a small network must predict the next latent (hidden) state from the current one and the next token, so a fixed-size state has to summarize the text so far (in latent space) (Teoh et al., 2026).



Tested on the BabyLM challenge, both changes help

- **Where we test it**: the BabyLM challenge, which is data and compute restricted. Models train on 10M words (the strict-small track) or 100M words (the strict track) of English; a child hears 2–7M words a year, so 100M words is a little more than the language input up to age 12 (Warstadt et al., 2023).



A child and four language models: Warstadt et al. (2023), Fig. 1.

- **What we find**: both changes help. The relational heads matter most for grammatical generalization; the auxiliary loss improves the fit to human reading times. On the official leaderboard (16 October 2026), our best 100M-word model ranks 6th of 55.

Roadmap

Relational attention

The Dual Attention Transformer

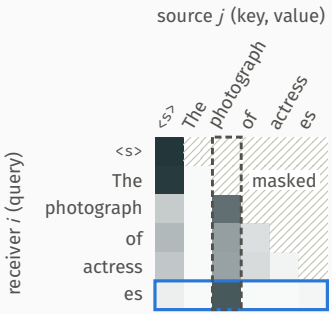
An RNN-like auxiliary loss: next-latent prediction

BabyLM: data, training, evaluation and results

Conclusion

Relational attention

Self-attention routes the features of the source it selects



One head of our 16-layer strict model (layer 14, head 4) on the BLiMP pair *The photograph of actresses does/*do hurt Cynthia.* (*actresses* is two tokens). Darker: larger weight; each row sums to 1.

- A language model predicts each token from the ones before it: in every layer, position i (the receiver) is updated from positions $j \leq i$ (its sources).
- The query–key product $\langle q_i, k_j \rangle$ (both rotated by position, RoPE) compares receiver x_i with source x_j: it encodes a relation between the two, and the softmax α_{ij} uses it to select sources. At *es*, where the verb is predicted, this head puts 0.83 on *photograph*, the noun the verb agrees with.
- The routed value $v_j = x_j W_v$ is a feature vector of the selected source; one computation mixes relational information (which source) and object-level information (what the source is).

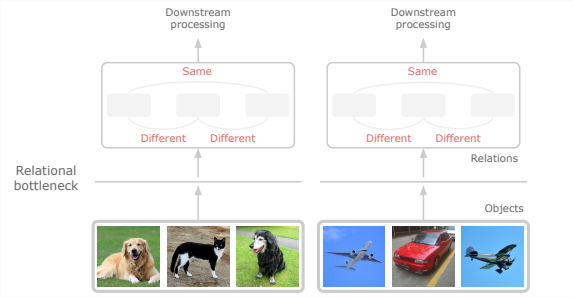
$$\mathbf{z}_i = \sum_{j \leq i} \underbrace{\alpha_{ij}}_{\text{selection}} \underbrace{v_j}_{\text{content}} \qquad (\mathbf{z}_i\text{: the attention output for receiver } i)$$

- Formal semantics divides this labor. Lexical semantics: word, object-like contributions to meaning (what *photograph* denotes). Compositional semantics: structure-based, relation-like ones (that *photograph* is the subject of the verb) (Montague 1973 a.o.)

Infants abstract relations from few examples; Transformers need many

- Seven-month-olds generalize algebraic patterns to new syllables (Marcus et al., 1999): familiarized with ABA triads such as *le di le*, *wi je wi*, they listen longer to *ba po po* (ABB) than to *ba po ba* (ABA). Few training examples, systematic transfer to out-of-training-sample (OOTS) objects.
- Self-attention Transformers need far more data for such rules, and their OOTS generalization stays poor (Webb et al., 2024; Kerg et al., 2022; Webb et al., 2021).
- Relational attention disentangled from self-attention greatly increases data efficiency and OOTS generalization on purely relational tasks (Altabaa et al., 2024; Altabaa and Lafferty, 2025).
- In language modeling, object-level and relational information must be integrated as well as disentangled; the evidence for DAT (the Dual Attention Transformer) as a language model is a modest perplexity gain on web text (Altabaa and Lafferty, 2025).
- We test whether the data efficiency transfers to language modeling, under the data & compute limits and broad evaluation of BabyLM (Choshen et al., 2026).

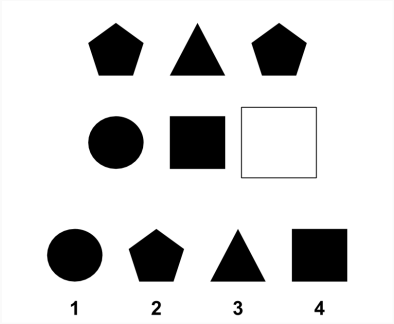
The relational bottleneck: later processing receives only relations



Webb et al. (2024), Fig. 1.

- Under the relational bottleneck, downstream processing receives the relations between inputs and none of their features (Webb et al., 2024).
- Dog, cat, dog and plane, car, plane then have the same downstream representation (ABA), so a pattern learned on one transfers to new objects.
- For a relational task, the relations by themselves suffice: restricting the representation to relations shrinks the space of possible representations, and that space still contains a sufficient one (Webb et al., 2024).

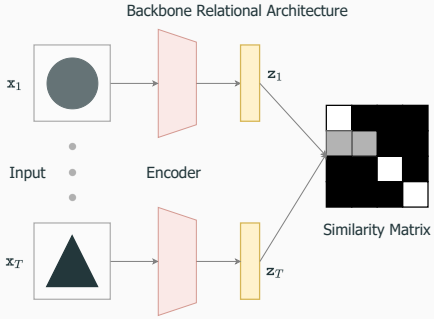
The identity rules task: find the rule in row 1, apply it to row 2



Webb et al. (2021), Fig. 1d.

- A visual version of the infant study of Marcus et al. (1999): row 1 follows a rule (here ABA); pick the object with which row 2 follows the same rule.
- Rules ABA, ABB or AAA; four choices, so chance is 25%. Here the answer is 1.
- 100 objects; m of them never appear in training, and every test problem is built from them. At $m = 95$ the model trains on 5 objects.
- The answer depends only on which positions contain the same object: a purely relational task.

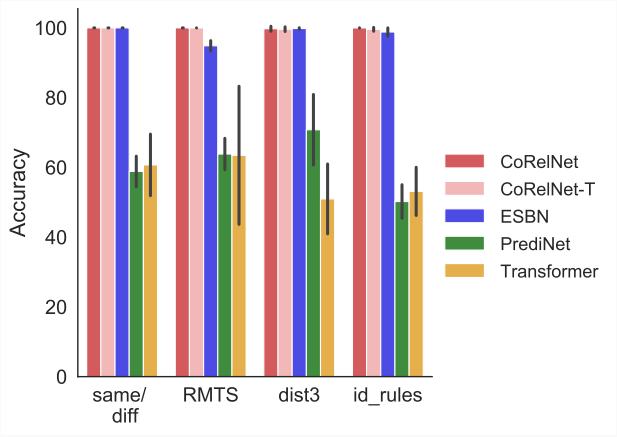
Relational models compute with dot products between objects



Kerg et al. (2022), Fig. 2 (top).

- ESBN, the emergent symbol binding network (Webb et al., 2021): an LSTM controller receives only learned key vectors (symbols), retrieved by dot-product matches between the current object and earlier ones.
- CoRelNet: encode the objects, compute all pairwise dot products (the similarity matrix) and pass only that matrix to an MLP (Kerg et al., 2022); CoRelNet-T, a Transformer layer instead of the MLP.
- An ABA row gives the same matrix for any A and B, so a rule read off the matrix holds for new objects.
- Adding the object encodings to the decoder input lowers accuracy on new objects (Kerg et al., 2022).

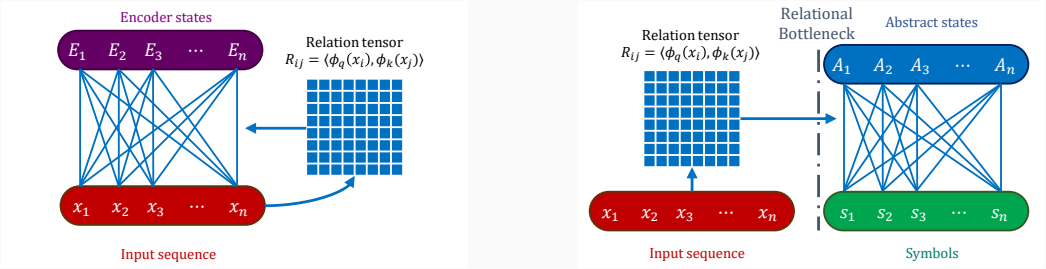
Models restricted to similarity scores generalize on all four relational tasks



Kerg et al. (2022), Fig. 3: most extreme regime of each task ($m = 95$; $m = 98$ for same/different); mean of 10 seeds.

- Four tasks from Webb et al. (2021): same/different, relational match-to-sample (RMTS), distribution of three, identity rules.
- CoRelNet, CoRelNet-T and ESBN compute their output from similarity scores only; on new objects their accuracy stays near 100%.
- The accuracy of the Transformer, which also receives the object features, is far lower, and so is that of PrediNet (Shanahan et al., 2020).

Relational cross-attention (RCA): attention with symbols as values



Altabaa et al. (2024), Fig. 1: (a) self-attention, (b) relational cross-attention. Red: object features; blue: relations; purple: mixed; green: symbols (their colours; in this talk blue is sensory, orange relational).

- Both use the relation matrix R_{ij}, the query–key products. Self-attention averages object features $\phi_v(x_j)$ (our v_j); relational cross-attention averages symbols s_j, so the objects enter only through R_{ij}.
- A symbol identifies an object (in the simplest scheme, by position) without encoding its features; the RCA heads of the Dual Attention Transformer (DAT) compute this average.

The Dual Attention Transformer

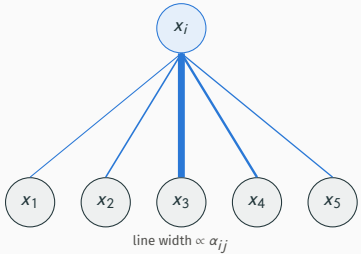
Sensory attention selects a source and routes its features

$$\mathrm{Attn}(x_i, \mathbf{x}) = \sum_{j \le i} \underbrace{\alpha_{ij}}_{\text{selection}} \underbrace{x_j W_v}_{\text{content}},$$

$$\alpha_i = \mathrm{Softmax}\big(\langle x_i W_q,\ x_j W_k \rangle\big)_{j \le i},$$

$\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^{n \times d}$: token vectors (the *objects*); x_i receiver.

- Sensory attention is standard self-attention; sensory b/c it routes the features of the source (Altabaa and Lafferty, 2025).
- W_q, W_k, W_v: learned query, key and value projections.
- α_{ij}: how strongly receiver i selects source j.
- The retrieved value $x_j W_v$ remains a sensory representation of the source object.



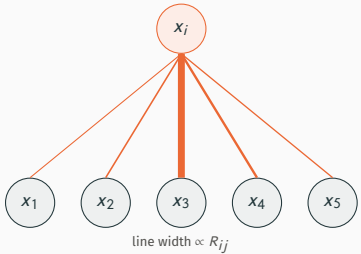
message from source j: $x_j W_v$
the features of source j

RCA: one score both selects the source and is the relation

$$\mathrm{RCA}(x_i, \mathbf{x}) = \sum_{j \le i} \underbrace{R_{ij}}_{\text{selection and relation}} \underbrace{s_{ij}\, W_s}_{\text{symbol}},$$

$$R_{ij} = \sigma_{\mathrm{rel}}\big(\langle x_i W_q,\ x_j W_k \rangle\big).$$

- Self-attention with symbols s_{ij} in place of the values (Altabaa et al., 2024): the features of j enter only through R_{ij}.
- R_{ij} is the weight of source j and the only relation passed on: one number per source and head.
- σ_{rel}: softmax in most Abstractor experiments; identity in our runs (sigmoid in the symbol comparison), so R_{ij} can be negative.
- Choosing a source and comparing with it can call for different criteria (Altabaa and Lafferty, 2025): at a verb, the subject is found by syntax and compared in number.



message from source j: $R_{ij}\ \ s_{ij} W_s$
one score, scaling the symbol for j

With RoPE, R_{ij} also depends on the distance $i - j$.

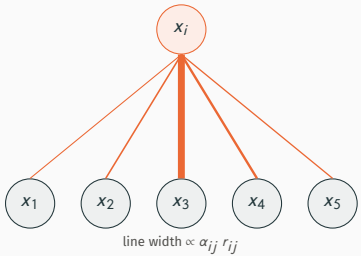
DisRCA: one score selects the source, a second score rates the relation

$$\mathrm{DisRCA}(x_i, \mathbf{x}) = \sum_{j \le i} \underbrace{\alpha_{ij}}_{\text{selection}} \underbrace{r_{ij}}_{\text{relation}} \underbrace{s_{ij}\, W_s}_{\text{symbol}},$$

$$\alpha_i = \mathrm{Softmax}\big(\langle x_i W_q^{\mathrm{attn}},\ x_j W_k^{\mathrm{attn}} \rangle\big)_{j \le i},$$

$$r_{ij} = \sigma_{\mathrm{rel}}\big(\langle x_i W_q^{\mathrm{rel}},\ x_j W_k^{\mathrm{rel}} \rangle\big).$$

- Two pairs of projections per head: one gives the selection weight α_{ij}, as in sensory attention (with RoPE); the other one relation score r_{ij} (no RoPE).
- Source j contributes in proportion to $\alpha_{ij}\, r_{ij}$, an AND of the two scores; only the symbol is routed, so weak selection with a strong relation and strong selection with a weak relation look alike.



message from source j: $\alpha_{ij}\, r_{ij}\ s_{ij} W_s$
two scores, scaling the symbol for j

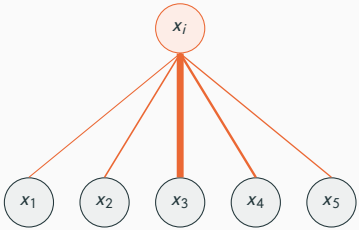
In the released code of Altabaa and Lafferty (2025), a step from RCA to relational attention; one of the three types we compare.

Relational attention routes a relation and a symbol for each source

$$\mathrm{RelAttn}(x_i, \mathbf{x}) = \sum_{j \le i} \underbrace{\alpha_{ij}}_{\text{selection}} \Big(\underbrace{r(x_i, x_j)\, W_r}_{\text{relation}} + \underbrace{s_{ij}\, W_s}_{\text{symbol}} \Big),$$

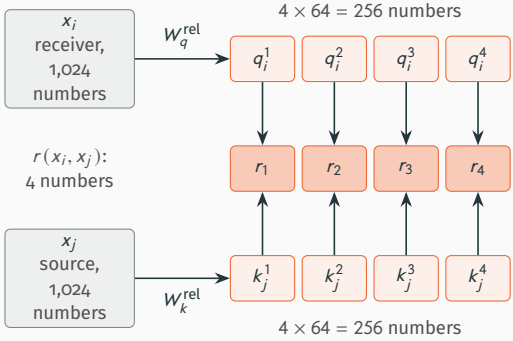
$$r(x_i, x_j) = \sigma_{\text{rel}}\Big(\big(\langle x_i W_q^{\text{rel},\ell},\ x_j W_k^{\text{rel},\ell} \rangle \big)_{\ell=1}^{d_r} \Big).$$

- $r(x_i, x_j) \in \mathbb{R}^{d_r}$: d_r learned comparison channels, each with its own projections; σ_{rel}: identity, sigmoid, tanh or softmax; separate query and key maps allow asymmetric relations.
- $s_{ij} \in \mathbb{R}^{d_s}$: a symbol, a vector tagging the source of the message.
- Unlike DisRCA, the relation is a vector and part of the message, not a factor of the weight.



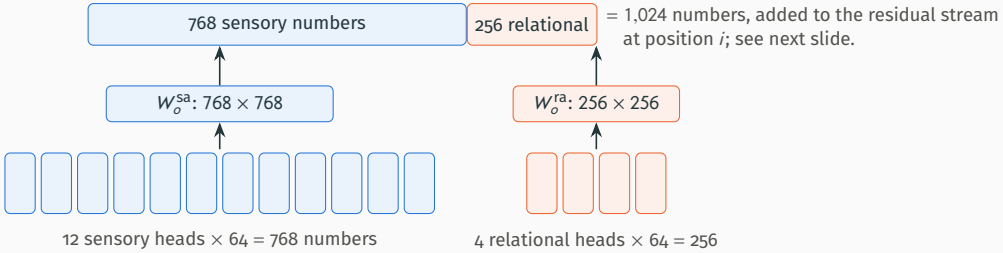
W_r, W_s map both into one space, where they add.

The relation vector: four channels, each comparing two 64-number projections



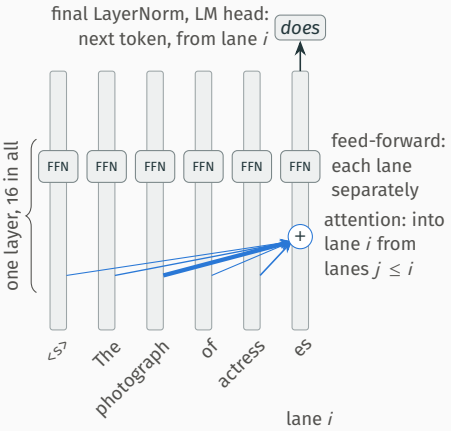
- Channel ℓ has a query map $W_q^{\text{rel},\ell}$ and a key map $W_k^{\text{rel},\ell}$, each $1{,}024 \times 64$; the 4 query maps form one $1{,}024 \times 256$ matrix W_q^{rel}, the 4 key maps another, W_k^{rel}.
- $r_\ell(x_i, x_j) = \langle x_i W_q^{\text{rel},\ell},\ x_j W_k^{\text{rel},\ell} \rangle / 8$, one number; $8 = \sqrt{64}$, as in self-attention; identity activation: r_ℓ can be any real number (either sign, any magnitude).
- no RoPE in $r_\ell(x_i, x_j)$, identity of a source encoded in the symbol s_{ij} (RoPE still in selection weight α_{ij}).
- Separate maps for receiver and source, so $r(x_i, x_j) \neq r(x_j, x_i)$ in general.
- One relation vector per receiver–source pair, computed once per layer and shared by the four relational heads; each head maps it with a separate W_r^h (4×64): the heads share one comparison and differ in the sources they select and the mix of channels they pass on (Altabaa and Lafferty, 2025).

The layer output: four relational head outputs next to twelve sensory ones



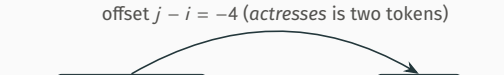
- Each group of heads has its own output map; the two results, side by side, are the 1,024 numbers added to the residual stream, on separate coordinates (768 sensory, 256 relational). The feed-forward layer that follows is applied to all 1,024.
- Parameters per layer: relational attention 1,377,280 (five $1{,}024 \times 256$ matrices: W_q, W_k, W_s, W_q^{rel}, W_k^{rel}; W_r, 1,024; W_o^{ra}, 65,536), sensory attention 2,949,120; no biases. Per head: 344,320 relational, 245,760 sensory.
- The symbol library is shared by all 16 layers; each layer has separate W_q^{rel} and W_k^{rel}, so every layer computes a new relation vector (as in the released code of Altabaa and Lafferty 2025).
- Layer 11 at *does*: length 2.10 for the 768 sensory numbers, 0.79 for the 256 relational numbers.

The residual stream: one lane per position, only attention crosses lanes



- **Residual stream** (Elhage et al., 2021): one 1,024-number vector, a lane in the stream, per position; lane i starts as the embedding of token i.
- Each layer adds to every lane twice (residual connections, He et al. 2016; pre-LayerNorm blocks): $x_i \leftarrow x_i + \text{Attn}_i(\text{LN}(x_1), \ldots, \text{LN}(x_i))$, then $x_i \leftarrow x_i + \text{FFN}(\text{LN}(x_i))$; lane i is the sum of the embedding and all outputs added so far.
- Attention is the only step that moves information between lanes: it adds to lane i a weighted sum computed from lanes $j \leq i$ (the causal mask). The LayerNorms, the output maps and the feed-forward layer act on each lane separately.
- In DAT, Attn is the sensory and the relational heads together. After the 16th layer, a final LayerNorm and the LM head turn lane i into a prediction of token $i+1$.

Illustration: a sensory and a relational message for subject–verb agreement



Sensory head at *shows*, selecting *photograph*

message: $x_{photograph}\, W_v$

the features of *photograph*: a singular noun, an image

Relational head at *shows*, selecting *photograph*

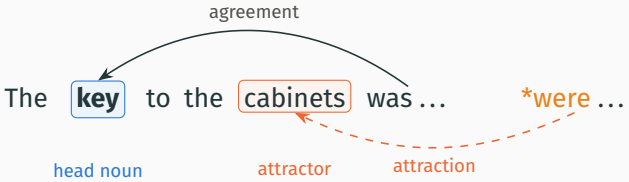
message: $r(x_{shows}, x_{photograph})\, W_r + s_{ij}\, W_s$

d_r comparison scores of *shows* with *photograph* (e.g. # match), plus symbol for 4 tokens back

- Any singular subject that compares with *shows* as *photograph* does, at the same distance, sends the same relational message; a rule learned from such messages applies to novel objects (Webb et al., 2024).
- The sensory message contains the meaning of *photograph*, and the verb has to suit it: *shows* suits an image, *laugh* a person.
- [Preview] DAT gains most on BLiMP island effects, NPI licensing and subject–verb agreement:

island effects	Who should Derek hug after shocking Richard?	*Who should Derek hug Richard after shocking?
NPI licensing	Only Amy ever won.	*Even Amy ever won.
subject–verb agreement	The sister of doctors writes.	*The sister of doctors write.

Agreement attraction: a verb agrees with a nearer noun instead of the subject



- In English, the verb agrees in number with the head noun of the subject: *key* in *the key to the cabinets*. The subject contains a second noun, nearer the verb and of the other number: the **attractor** *cabinets*.
- Bock and Miller (1991): speakers heard a sentence beginning such as *The key to the cabinets*, repeated it and completed it into a sentence. With a singular head noun and a plural attractor, 50 of 320 completions had a plural verb (*The key to the cabinets were …*).
- With a plural head noun and a singular attractor (*The keys to the cabinet*), 7 of 320 had a singular verb: plural, the marked number, attracts far more often.

In reading, an ungrammatical plural verb is easier after a plural attractor

	attractor	reading at the verb
The **slogan** on the poster unsurprisingly **was** designed …	singular	no slowdown
The **slogan** on the posters unsurprisingly **was** designed …	plural	no slowdown
*The **slogan** on the poster unsurprisingly were designed …	singular	slower
*The **slogan** on the posters unsurprisingly were designed …	plural	**less slow**: attraction

Items of Wagers et al. (2009), Experiment 4, as listed by von der Malsburg and Padó (2026).

- Self-paced reading: readers press a key for each next word, and the time per word is longer where a sentence is hard to read.
- An ungrammatical verb (*were* after *slogan*) slows reading down; after the plural *posters*, the slowdown is smaller (Pearlmutter et al., 1999; Wagers et al., 2009).
- Attraction is asymmetric: it appears with a singular subject and a plural attractor, not the reverse (Pearlmutter et al., 1999); also in relative clauses (Wagers et al. 2009, Exp. 3).
- Is agreement attraction present in our trained models at all, and do relational and sensory attention have a differential effect on it? Some early results below.

An RNN-like auxiliary loss: next-latent prediction

A Transformer attends to every earlier token; an RNN keeps one state

- In a Transformer, memory grows with the sequence: attention can retrieve any earlier token (in the context window). An RNN carries one fixed-size state forward, updated in the same way at every token: the entire past is compressed into it.
- Teoh et al. (2026) argue that simpler, more compact explanations of the training data generalize better, and that Transformers, which can look up any earlier token, often learn shortcuts that do not.
- Human comprehension works under a Now-or-Never bottleneck (Christiansen and Chater, 2016): input must be chunked and compressed into hierarchical representations before sensory memory decays. This eager incremental processing underlies expectations in rational parsing (Hale, 2011) and cue-based memory retrieval (Lewis and Vasishth, 2005), the favored account of agreement attraction (Wagers et al., 2009).
- With NextLat, a Transformer is trained under the compression pressure of an RNN; the architecture and inference stay unchanged.

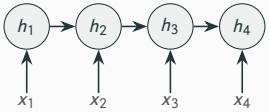
With NextLat, hidden states are trained to act like an RNN state

Transformer



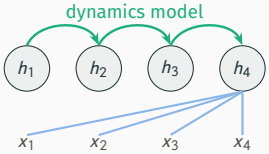
h_t attends to every earlier token

RNN



one state: $h_{t+1} = f(h_t, x_{t+1})$

Transformer + NextLat



dynamics model: $h_t, x_{t+1} \mapsto \hat{h}_{t+1}$

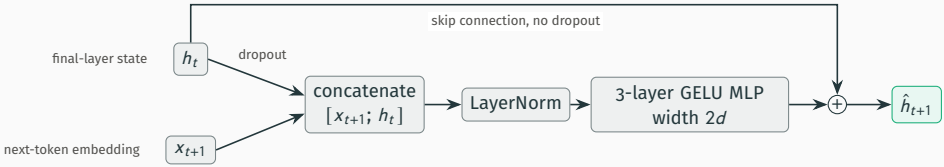
- To predict h_{t+1} from h_t and x_{t+1}, h_t must already contain what h_{t+1} contains about the past: each fixed-size state is pushed to summarize the history.
- The constraint applies only in training: attention still covers the whole history, and the dynamics model, a small network, is not used at inference.
- Training loss = $\mathcal{L}_{\text{NTP}} + \mathcal{L}_h + 0.5\,\mathcal{L}_{\text{KL}}$, NextLat horizon 1 (our weights, Teoh et al. (2026) tune them per domain, the KL weight from 0 to 1, and predict up to 8 steps ahead):

  $\mathcal{L}_{\text{NTP}}$: the next-token cross-entropy $\mathcal{L}_h$: Smooth L1 loss between $\hat{h}_{t+1}$ and h_{t+1}

  $\mathcal{L}_{\text{KL}}$: KL divergence between the next-token distributions of the LM head from $\hat{h}_{t+1}$ and from h_{t+1}

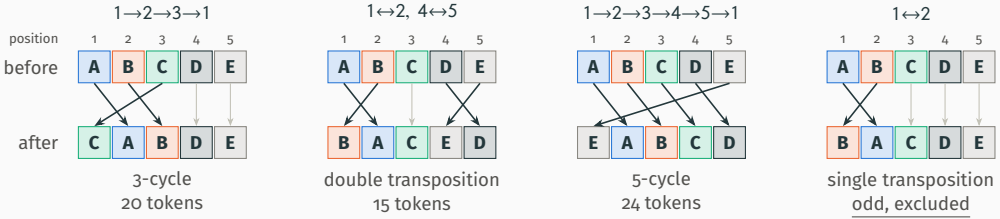
  (no gradient flows into h_{t+1} in $\mathcal{L}_h$ or $\mathcal{L}_{\text{KL}}$, nor into the LM head in $\mathcal{L}_{\text{KL}}$; their gradients flow into DAT through h_t and x_{t+1})

The dynamics model in our DAT-based LM predicts h_{t+1} from h_t and the next token



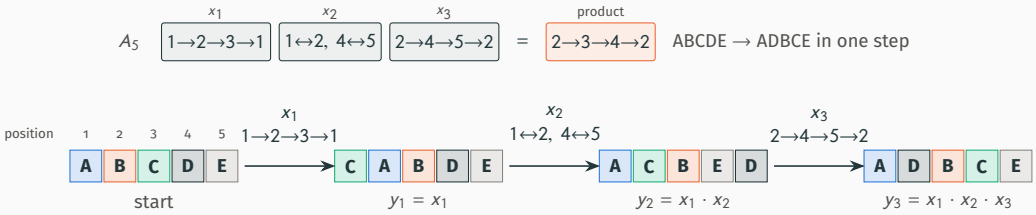
- The dynamics model computes $\hat{h}_{t+1} = h_t + \mathrm{MLP}(\mathrm{LN}([x_{t+1}; h_t]))$: the MLP outputs the change to the state due to token x_{t+1}, and the skip from h_t is the only residual connection.
- LN: LayerNorm over $[x_{t+1}; h_t]$; MLP: three layers of width $2d$ with GELU (Gaussian error linear unit) activations. Dropout (0.1, as elsewhere in the model) applies to h_t entering the LayerNorm, and h_t enters the skip path without dropout.
- The dynamics model is a small network compared to the DAT backbone: it adds ≈5.9M parameters at strict-small scale, ≈10.5M for our 16-layer strict model, and is discarded after training.

The A_5 word problem: each token is one of 60 permutations of five objects



- One of the motivations for NextLat in Teoh et al. (2026): the A_5 word problem (synthetic task).
- Five objects, A to E, in positions 1 to 5. Each token is a permutation of the objects: 1→2→3→1 moves $A \to 2, B \to 3, C \to 1$; D and E stay (grey arrows).
- The tokens are the even permutations: every way of writing one as transpositions (exchanges of two positions) uses an even number. 1→2→3→1 is positions 1,2 then 1,3 exchanged (ABCDE → BACDE → CABDE), or 2,3 then 1,2 (ABCDE → ACBDE → CABDE): two ways of writing the same token.
- 60 of the 120 permutations are even: 1 identity, 20 3-cycles, 15 double transpositions, 24 5-cycles.
- Chaining two tokens gives a token, chaining (product) is associative, the identity is a token, and each token is undone by a token, its inverse (1→2→3→1 by 1→3→2→1): the 60 tokens form a group.

The word problem: a sequence of tokens has a product, one of the 60 tokens



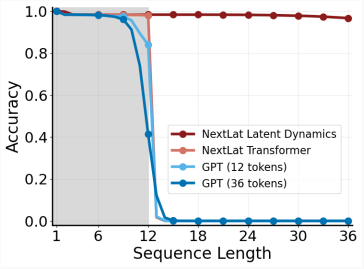
- A word is defined as a finite sequence of elements of a group, written in a row like the letters of a string (Merrill et al., 2024, Section 3.1). For A_5 the letters are the 60 tokens: $x_1 x_2 x_3$ above is a word of length 3.
- The product of a word $x_1 x_2 \cdots x_t$ is the single element y_t the whole sequence amounts to: applied in order, the three tokens $x_1 x_2 x_3$ turn ABCDE into ADBCE, and so does the one token 2→3→4→2, their product.
- The word problem: given a word $x_1 x_2 \cdots x_t$ of any length, find its product $y_t = x_1 \cdot x_2 \cdot \ldots \cdot x_t$ (Merrill et al., 2024, Definition 3.1); y_t is always one of the 60 tokens.

The dynamics model by itself stays above 95% up to 36 tokens

Transformer or RNN: two ways to compute every y_t

Transformer: recompute. At position t, attention sees all t tokens, but applying them in order takes about $\log_2 t$ rounds of pairwise composition, a few layers each (Liu et al., 2023). The order matters ($x_1 \cdot x_2$ makes ACBED from ABCDE, $x_2 \cdot x_1$ makes CBAED), and the number of layers needed grows with the length (Merrill et al., 2024).

RNN: carry forward. The state after t tokens can be the intermediate product y_t; each step applies one token to it, $y_t = y_{t-1} \cdot x_t$. The update is the same at every position, so an update learned on 12 tokens applies unchanged for 36.



Teoh et al. (2026), Fig. 11: length generalization for the A_5 word problem.

- Horizontal: sequence length l, 1 to 36; vertical: accuracy, the share of rows with all of the first l outputs correct; grey: up to 12, the training length.
- Up to 12 tokens, the NextLat Transformer (light red) stays near 1; GPT trained on 12 tokens (light blue) falls below it before 12.
- Beyond 12 tokens both Transformers fall to about 0. GPT trained on 36-token rows (dark blue), the same two-layer model with the same context window, falls even earlier, before 12 tokens: the limit lies in the depth (Merrill et al., 2024).
- The dynamics model (dark red; 2.62M parameters, the Transformer 6.43M), started from the Transformer state of the first token, stays above 95% up to 36 tokens.
- In our BabyLM models, the dynamics model (our own reimplementation) serves only in training; we evaluate only the Transformer / DAT.

Interim Summary: Relational BabyLM – two inductive biases in one decoder

Architecture: DAT

Our reimplementation of the Dual Attention Transformer (Altabaa and Lafferty, 2025): sensory self-attention plus a parallel *relational attention*, which selects sources as usual but retrieves a pairwise relation vector and an abstract symbol (a tag identifying the source).

Separate streams for object information (lexical semantics: what words contribute to meaning) and relational information (compositional semantics: what syntactic structure contributes to meaning).

Training: an auxiliary NextLat loss

A small network, the dynamics model, is trained to compute the next hidden state from the current one and the next token, as an RNN updates its state (Teoh et al., 2026).

Each state must then contain what the past contributes to predicting the future, like an information state in dynamic semantics (Kamp, 1981; Heim, 1982; Groenendijk and Stokhof, 1991). The dynamics model is discarded after training.

- Two bottlenecks: relational heads pass on only relations and source tags (Webb et al., 2024); with NextLat, each state is trained to summarize the past.
- We submit models to both BabyLM tracks (strict-small, 10M words & strict, 100M); multi-seed strict-small experiments separate the effects of the two changes.

BabyLM: data, training, evaluation and results

BabyLM 2026 training data: six English sources, 10M or 100M words

Source	What it is	Strict-small	Strict
CHILDES	conversations of young children and caregivers (MacWhinney, 2000), one utterance per line	2,841,101	28,410,878
Gutenberg	public-domain books, one paragraph per line	2,557,721	25,576,896
OpenSubtitles	movie and TV subtitles	2,282,875	22,828,736
Simple Wikipedia	encyclopedia articles in Simple English	1,531,435	15,314,315
BNC spoken	transcribed British speech, one turn per line	762,073	7,620,671
Switchboard	transcribed telephone conversations	24,791	248,491
Total words		9,999,996	99,999,987

- A word is a whitespace-separated string, so CHILDES speaker codes such as `*MOT:` count (about 17% of the CHILDES words).
- The 2026 release is detoxified: hate-speech, sentiment and emotion classifiers flag sentences, and the flagged ones are filtered with gender and race word lists and a slur lexicon (Choshen et al., 2026).
- All our controlled experiments, on architecture, symbol mechanism and head ratio, are run on strict-small (10M words).

Examples of training text

CHILDES

```
*COL: yes your lunch.
*CHI: more.
*MOT: more what?
*CHI: more tapioca.
```

Switchboard

```
B: Yeah,
B: I'm in Texas.
B: Where are you at?
```

BNC spoken

```
Why have you just discounted it to there?
Sorry?
```

OpenSubtitles

```
It's a lovely home.
- And?
You're about to get 86'd.
```

Simple Wikipedia

```
= = = Chairman District Council Faisalabad = = =
Chairman District Council Faisalabad is the
deputy mayor who heads the Faisalabad Municipal ...
```

Gutenberg (one paragraph is one line)

```
"Don't do too much," Tom said.  "Remember you
had a pretty tough time yesterday."
```

Long lines are wrapped on this slide for display. Lines of the form `= = = ... = = =` separate documents: CHILDES transcripts, Gutenberg books, Wikipedia articles.

Training requirements: word budget, 10 epochs, checkpoints by words seen

- **Word budget**: at most 100M words (strict) or 10M words (strict-small) of training text. Text used to train any learned component, a tokenizer included, counts toward the budget (Choshen et al., 2026).
- **Exposure**: at most 10 epochs over the training data, so a strict-small model is trained on at most 100M words and a strict model on 1B. We train every model for 10 epochs.
- **Checkpoints**: after 1M, 2M, …, 10M words seen, then 20M, …, 100M, and for strict 200M, …, 1,000M. Each checkpoint is scored on a fast subset of the evaluation, and the age-of-acquisition measure is computed from these learning curves; the final model is scored on the full evaluation, finetuning included.

Our models vary in activation, head split, objective and head tying

- Feed-forward activation: GELU (Gaussian error linear unit; Hendrycks and Gimpel 2016) or SwiGLU (Shazeer, 2020), a gated linear unit (Dauphin et al., 2017) with the Swish activation (Ramachandran et al., 2017).
- Head split: the attention heads of DAT are n_h^{sa} sensory and n_h^{ra} relational heads, written SA/RA (relational heads of any type: RCA, DisRCA or RA): half to each stream in 6SA/6RA, three quarters to sensory attention in 9SA/3RA. Every submitted model has the 3:1 allocation (12SA/4RA in the 16-layer, 1,024-dimensional model).
- Objective: NTP (next-token prediction) ± NextLat. The LM head is tied (it reuses the token-embedding matrix) or untied (its own matrix) in NTP (no NextLat); NextLat runs are untied, as in Teoh et al. (2026).

The BabyLM 2026 evaluation suite has three families of measures

Zero-shot ranking

No parameter updates.

BLiMP, BLiMP supplement (Warstadt et al., 2020)
COMPS (Misra et al., 2023)
Entity tracking (Kim and Schuster, 2023)
EWoK (Ivanova et al., 2025)
Global PIQA (Chang et al., 2025)

Finetuning

Seven (Super)GLUE tasks:

BoolQ, MNLI, MRPC, MultiRC, QQP, RTE, WSC

Human-likeness

Reading: fit of model surprisal to eye-tracking and self-paced reading times (de Varda et al., 2023)

Age of acquisition (AoA) (Chang and Bergen, 2022)

- Leaderboard aggregates: **NLP**, the average of seven components (the six zero-shot scores and one (Super)GLUE score, itself the average of the seven finetuned tasks); **Human-like**, the average of reading and AoA; **Overall**, the average of all nine.

Zero-shot benchmarks: what each one tests

Benchmark	What it tests	Items scored
BLiMP	67 minimal-pair paradigms: agreement, islands, filler–gap dependencies, binding, NPI licensing, quantifiers, …	59,875
BLiMP supplement	hypernymy, question–answer congruence, subject–auxiliary inversion, turn-taking	5,218
EWoK	world knowledge in 11 domains: with which of two contexts a fixed target sentence is more plausible	7,618
COMPS	property attribution and inheritance through novel concepts, with distractors	91,028
Entity tracking	final contents of boxes after a sequence of operations (five options)	6,780
Global PIQA	physical commonsense, English: parallel (4 options) and non-parallel (2 options) splits, score = mean of the two	103 + 100

- 89.4% of the 67,000 published BLiMP pairs remain after the BabyLM vocabulary filtering; the BLiMP score is the unweighted mean over paradigms.

Zero-shot evaluation: which of two strings gets the higher log-probability?

$$\log p(x_1 \dots x_T) = \sum_{t=1}^{T} \log p(x_t \mid \texttt{<s>}, x_1, \dots, x_{t-1})$$

Ann	sees	Tanya	.	$\log p(\text{good})$
Ann	see	Tanya	.	$\log p(\text{bad})$

- The item is correct when $\log p(\text{good}) > \log p(\text{bad})$. No training on the task: one forward pass per string.
- The term for *Ann* is the same in both sums, so the comparison depends on $p(\textit{sees} \mid \textit{Ann})$ against $p(\textit{see} \mid \textit{Ann})$ and on the words after them.
- In BLiMP, whole sentences are scored; in EWoK, COMPS, entity tracking and Global PIQA, only the candidate continuation after a prefix. Ties are broken at random.

BLiMP: 67 paradigms of minimal pairs

Phenomenon	Acceptable	Unacceptable
Subject–verb agreement with a noun in between	Ann sees Tanya. The sister of doctors writes.	*Ann see Tanya. *The sister of doctors write.
Determiner–noun agreement	Tanya fixed this cup.	*Tanya fixed this cups.
Island (adjunct)	Who should Derek hug after shocking Richard?	*Who should Derek hug Richard after shocking?
Filler–gap dependency	Colleen knew the lady that Frank loved.	*Colleen knew who Frank loved the lady.
NPI licensing	Only Amy ever won. Did Jill ever die?	*Even Amy ever won. *Jill did ever die.

- Each paradigm has 1,000 pairs generated from a grammar (Warstadt et al., 2020); after the BabyLM vocabulary filtering, 59,875 of the 67,000 pairs remain (89.4%).
- The score is the unweighted mean of the 67 paradigm accuracies; chance is 50%.

Agreement attraction: the heads that track subjects are sensory

- **How we find the heads that track subjects.** In a treebank with hand-annotated subjects (UD English-EWT, Silveira et al., 2014) we take the 10,235 subjects that are not the word right before their predicate. A head finds such a subject when, at the later of the two words, its largest attention weight is on the other one. A head that always attended two words back, their most common position, would find 42.8% of them; following Ryu and Lewis (2025), a head is kept if it finds at least 10% more: $1.1 \times 42.8\% = 47.1\%$. Leaving out the subjects right before the predicate keeps out heads that only attend to the previous word. **8 of the 192 sensory heads pass**, one in each of layers 3 to 10; **none of the 64 relational heads**.
- **The test:** 543 BLiMP pairs such as *A picture of many bikes has / *have impressed Grace*, where the ungrammatical verb agrees with the attractor *bikes* (the grammatical sentence is the more probable in 501). We compare the attention weights at the verb in the two sentences.
- **With the attracting verb, all 8 heads attend less to the head noun and more to the attractor**: on average 0.235 → 0.165 on the head noun, 0.095 → 0.128 on the attractor. Head 5 of layer 8: 0.302 → 0.093 and 0.018 → 0.088; the attractor gains in 96.5% of the pairs and in all 42 where the ungrammatical sentence is the more probable.
- **The weights are more spread out**, as Ryu and Lewis (2025) predict for interference: entropy 1.737 → 1.786 bits on average ($t(542) = 13.08$, $p < 0.001$). In layers 7 to 10 the weight leaves the head noun and spreads; in layers 3 to 6 it concentrates on the attractor.

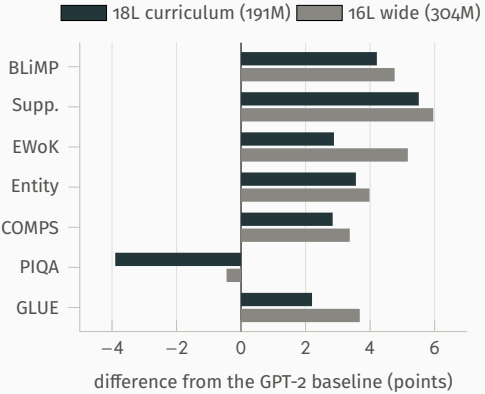
On the strict track (100M words), our best model ranks 6th of 55

Main result

Model	BLiMP	Supp.	EWoK	Ent. trk.	COMPS	PIQA	GLUE	Reading	AoA	Overall	NLP
GPT-2 baseline (12L)	74.73	65.00	54.37	16.91	55.85	36.62	67.75	6.93	−11.58	40.73	53.03
DAT 12L (NTP)	79.81	60.03	56.66	20.84	57.84	**37.65**	65.92	5.76	−12.02	41.39	54.11
DAT 18L (NextLat)	**80.62**	62.01	56.99	**20.93**	58.36	34.72	68.90	6.49	0.00	43.23	54.65
DAT 18L (NextLat, curric.)	78.94	70.51	57.25	20.47	58.69	32.72	69.95	6.22	0.00	43.86	55.50
DAT 16L wide (NextLat, curric.)	79.49	**70.96**	**59.54**	20.89	**59.22**	36.17	**71.43**	**7.10**	−9.48	**43.92**	**56.81**

- Official leaderboard, 16 October 2026 (55 entries): the 16L wide model ranks 6th overall and 3rd on the NLP-task subset; the 18L curriculum model ranks 7th. The EWoK score of the 16L wide model (59.54) is the highest on the strict track.
- All DAT entries: RoPE-based relative symbols, SwiGLU, Muon/LambW. 12L: 123M parameters (RCA); 18L: 191M (RCA); 16L wide: 304M (RA with four relation channels).
- The GPT-2 baseline (98M parameters, AdamW) comparison is not parameter- or training-recipe-matched.

Our two strongest models are above the GPT-2 baseline on most benchmarks



Not parameter- or training-recipe-matched: the baseline has 98M parameters trained with AdamW; ours 123M–304M with Muon/LambW.

- Each bar: our model minus the baseline, in points; e.g. BLiMP, 16L wide: 79.49 vs. 74.73. Reading and AoA (on other scales) are left out of the chart.
- 16L wide: higher on 8 of 9 benchmarks (all but Global PIQA); 18L curriculum: 7 of 9 (all but Global PIQA and reading).
- AoA: 0.00 for the 18L model, i.e. no reliable correlation; the baseline score, −11.58, is significantly negative.
- With the curriculum (Phase A line EOS, Phase B document EOS), the 18L model is 8.5 supplement points higher than without it (70.51 vs. 62.01) and 1.68 BLiMP points lower (78.94 vs. 80.62).

The two strongest strict-track models

	18L curriculum model	16L wide model
Leaderboard name	DAT Strict Curriculum	DAT Strict NextLat Final
Parameters	191.2M	304.3M
Layers / hidden dimension	18 / 768	16 / 1,024
Heads (SA/RA), attention type	9/3, RCA	12/4, RA (4 relation channels)
Feed-forward	SwiGLU, 3,072	SwiGLU, 4,096
Tokenizer (GPT-BERT, 100M words)	official 2025 baseline	2026 retraining
Batch	48 sequences of 512 tokens	36 sequences of 512 tokens
Curriculum (line-EOS / document epochs)	6 / 4	2 / 8
Muon lr, Phase A / B	0.02 / 0.008	0.02 / 0.005
LambW lr, Phase A / B	10^{-3} / 4×10^{-4}	10^{-3} / 2.5×10^{-4}
Weight decay	0.01	0.05
Submitted weights	raw final checkpoint	EMA (decay 0.999)

- Shared: RoPE-based relative symbols, untied head, NextLat (horizon 1), seed 1, 10 epochs.
- Phase A: EOS after every line (utterance); Phase B: document EOS, from the Phase-A EMA.

Two optimization recipes: AdamW for strict-small, Muon with LambW for strict

Strict-small analyses (140 runs)

AdamW (Loshchilov and Hutter, 2019) for every parameter: β_1=0.9, β_2=0.999, no weight decay; learning rate 5×10^{-5} (architecture comparison, head-ratio sweep) or 10^{-3} (symbol retrieval).

Strict-track submissions

Muon (Jordan et al., 2024) for the hidden weight matrices (88.9% of the parameters of the 16-layer model); **LambW**, our variant of LAMB (You et al., 2020), for embeddings, LM head, norms and biases. Weight decay 0.01 or 0.05; 1% warmup, then cosine decay to 10% of the initial rate.

- At 12 layers on the strict track, the Muon/LambW recipe is better than AdamW (lr 5×10^{-4}) on all five accuracy measures (NTP models: BLiMP +1.73, supplement +1.15, EWoK +1.77, entity tracking +1.99, COMPS +1.02 points); reading measures decrease slightly.

  The comparison changes optimizer, learning rate and weight decay together; one seed only.

NextLat improves 5 of 7 finetuned (Super)GLUE accuracies

- Two 12-layer strict-track models trained with Muon/LambW: NTP with a tied head, NextLat with an untied head.
- The seven tasks (Wang et al., 2018, 2019): BoolQ, yes/no questions about a passage; MultiRC, whether an answer to a question about a paragraph is correct; WSC, what a pronoun refers to; MNLI and RTE, whether a premise entails a hypothesis; MRPC and QQP, whether two sentences (QQP: two questions) mean the same.
- NextLat is higher on MultiRC (+5.7 points), WSC (+3.8, two of the 52 WSC validation items), RTE (+2.9), MRPC (+1.5) and QQP (+1.1); the two are equal on BoolQ (68.4%), and NextLat is 0.2 points lower on MNLI.
- Two-sided paired t-test across the seven task accuracies: $t(6) = 2.61$, $p = 0.040$.
- All seven values are accuracies; on the leaderboard, MRPC and QQP are scored by F1, so these values differ from the leaderboard entries.
- The two models differ in LM-head tying and parameter count ($\approx$ 12.6M).

Strict-small (10M words): our NextLat model ranks 8th on BLiMP

Model	BLiMP	Supp.	EWoK	Ent. trk.	COMPS	PIQA	GLUE	Reading	AoA	Overall
GPT-2 baseline (12L)	65.23	57.25	50.63	**19.10**	51.81	35.09	63.80	5.63	−12.15	37.38
DAT 12L (NTP, tied, 123M)	70.40	**57.94**	50.25	18.62	**52.79**	**35.64**	62.96	6.56	−14.00	37.91
DAT 12L (NextLat, untied, 136M)	**72.34**	56.40	**52.93**	18.45	52.69	33.65	**63.97**	**7.12**	−14.15	**38.16**

- Official leaderboard, 16 October 2026 (128 entries): 62nd (NTP) and 55th (NextLat) by Overall average. The NextLat model ranks 8th on BLiMP (72.34 vs. 65.23 for the baseline) and 6th on EWoK (52.93). In the paper, we report an earlier snapshot: 60th and 53rd of 127.
- NextLat vs. NTP: BLiMP +1.9, EWoK +2.7, reading 7.12 vs. 6.56, (Super)GLUE 63.97 vs. 62.96. The two also differ in LM-head tying and parameter count. In the controlled runs, the objective has no significant BLiMP effect within DAT ($\chi^2(1) = 1.53$, $p = 0.216$).
- Both: 9SA/3RA RCA, SwiGLU, RoPE-based relative symbols, Muon/LambW, 10 epochs. The GPT-2 baseline comparison is not parameter- or training-recipe-matched.

Strict-small models share one backbone

- 12 layers, 768 hidden dim ('d-model'), BPE vocabulary 16,384, RoPE (queries and keys rotated by position in every attention layer), pre-LayerNorm blocks, GELU or SwiGLU feed-forward activations.
- Standard Transformer baseline: 12 self-attention heads. DAT replaces them with 12 heads, n_h^{sa} sensory and n_h^{ra} relational (SA/RA).
- Parameters: tied ≈123.3M, untied ≈135.9M (the untied head adds ≈12.6M). The NextLat dynamics model adds ≈5.9M during training only.
- 10 epochs, gradient clipping at 1.0.

Two types of statistical models, both fit on strict-small

Binomial generalized linear mixed models (GLMMs) on BLiMP pseudo-items

- Pseudo-item: one of 100 fixed bins of the items of a subtest, scored as correct out of items.
- Baseline-vs-DAT: standard Transformer vs. 9SA/3RA RCA/SwiGLU DAT; fixed effects for architecture, LM-head tying, NextLat objective.
- Internal-DAT: adds attention type, head split, activation and symbol position, context length.
- Each factor is tested with a Type III Wald χ^2 test (sum-to-zero coding); pairwise contrasts are Tukey-adjusted; experiment, BLiMP subtest and pseudo-item have crossed random intercepts.

Reading regression (linear mixed model)

Response: per-experiment ΔR^2 from adding surprisal, for eleven measures (four eye-tracking, self-paced reading, six ERP components), in base and spillover regressions. Random intercepts for experiment and measure; fixed effects for objective, model type, tying, context length and regression condition.

Our experiments: two complete comparisons and 77 exploratory runs

Symbol retrieval: 35 runs

7 conditions $\times$ 5 seeds; RCA, SwiGLU, 264 tokens.

BLiMP: 69.34% with learned relative symbols, lower only with the two relational-symbolic conditions (66.32%, 68.33%). RoPE-based relative symbols: 69.43%, no library parameters.

SA/RA head ratio: 28 runs

7 ratios $\times$ 3 seeds at 512 tokens, plus one 264-token run per ratio; RCA, SwiGLU.

A higher relational fraction lowers BLiMP ($t = -5.887$, $p < 0.001$) and EWoK ($t = -2.323$, $p = 0.021$); BLiMP peaks at 6SA/6RA (70.33%).

The other 77 runs are exploratory / less systematic: partially crossed, with as many seeds as needed to fix part of the architecture, a hyperparameter or part of the training pipeline. All 140 runs: strict-small, 10M words.
SwiGLU chosen over GELU in early comparison, kept in all submitted models and the 2 experiments above (35 + 28 runs).

Main results of the exploratory runs:

- DAT is above the standard Transformer on BLiMP in every observed tying $\times$ objective cell (best 70.66% vs. 68.16%; $\chi^2(1) = 139.43$, $p < 0.001$), and on entity tracking, by 3.82 to 7.96 points
- NextLat models fit human reading measures better ($t = 5.22$, $p < 0.001$)
- RA, RCA and DisRCA do not differ significantly on BLiMP at 10M words ($\chi^2(2) = 3.50$, $p = 0.174$); at 100M words, in one matched 18-layer pair (one seed each), RA is higher than RCA on six of seven measures

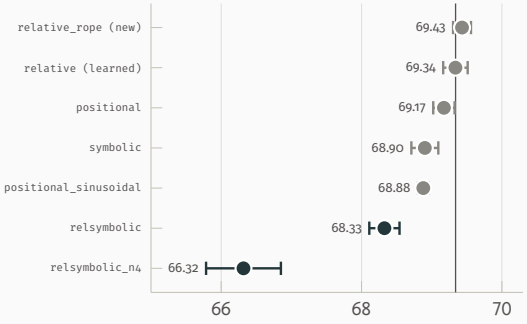
Seven symbol mechanisms in the controlled comparison

Condition	Symbol for source j at receiver i	Library
`relative`	indexed by the offset $j - i$	learned
`relative_rope` (new)	RoPE applied to the offset $j - i$	none
`positional`	indexed by the absolute position j	learned
`positional_sinusoidal`	absolute position j, sinusoidal encoding (Altabaa et al., 2024)	fixed
`symbolic`	convex combination of library symbols, selected by learned feature templates	learned
`relsymbolic`	symbolic attention over a local relation profile, causal neighborhood $k = 2$	learned
`relsymbolic_n4`	the same with neighborhood $k = 4$	learned

- `relative_rope`, RoPE-based relative symbols (new): one fixed vector, rotated by the offset $j - i$, as RoPE rotates queries and keys (Su et al., 2023); no learned parameters, and nearby offsets get similar symbols.
- Five seeds per condition, RCA relational attention with SwiGLU, 264-token context.

BLiMP is lower only with the relational-symbolic mechanisms

Main result

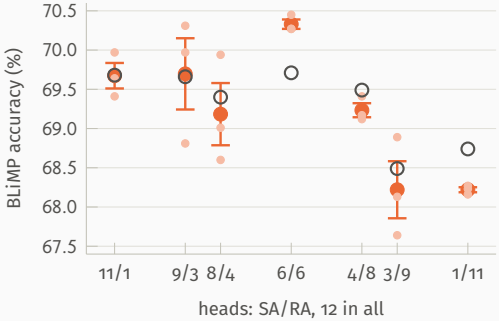


Vertical line: learned relative symbols (69.34). Dark: significantly lower; grey: not significantly different. Error bars: ±1 standard error of the mean over the 5 seeds, how much the mean varies from one set of 5 seeds to another.

- 7 conditions × 5 seeds (RCA, SwiGLU, 264 tokens); GLMM on BLiMP subtests: $\chi^2(6) = 134.29$, $p < 0.001$.
- Against learned relative symbols (69.34%): `relsymbolic_n4` 66.32% ($z = 9.37$, $p < 0.001$), `relsymbolic` 68.33% ($z = 3.21$, $p = 0.023$).
- The other five do not differ significantly (all pairwise $p > 0.5$, Tukey).
- RoPE-based relative symbols: 69.43% vs. 69.34%, with no learned parameters.

A larger relational stream lowers BLiMP

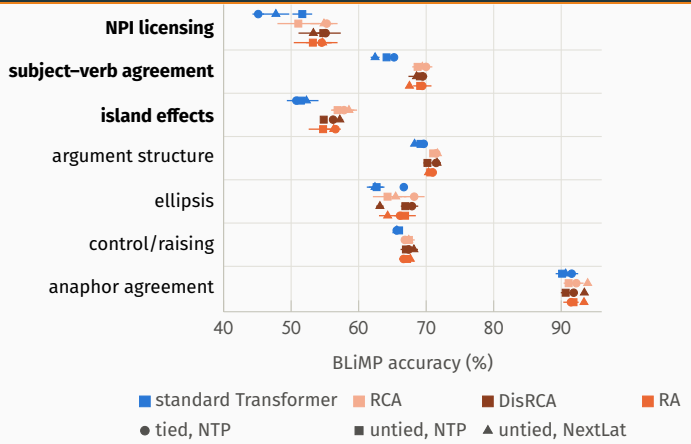
Main result



x: the SA/RA split, spaced by the number of relational heads; *y*: BLiMP accuracy (%). Error bars: ±1 standard error of the mean over the 3 seeds; e.g. 6SA/6RA: seeds 70.27, 70.45, 70.27%, mean 70.33%, standard error 0.06.

- Seven SA/RA splits of 12 heads (RCA, SwiGLU, learned relative symbols); 3 seeds at 512 tokens.
- A higher RA fraction lowers BLiMP (slope -1.827, $t = -5.887$, $p < 0.001$) and EWoK (-1.039, $t = -2.323$, $p = 0.021$); no significant effect on COMPS ($t = -1.287$, $p = 0.201$) or reading ($t = 0.710$, $p = 0.488$).
- With 12 heads in all, more relational heads means fewer sensory ones.
- BLiMP peaks at 6SA/6RA (70.33%); the submitted 9SA/3RA is 0.6 BLiMP points lower, with +1.4 supplement points and the best COMPS.

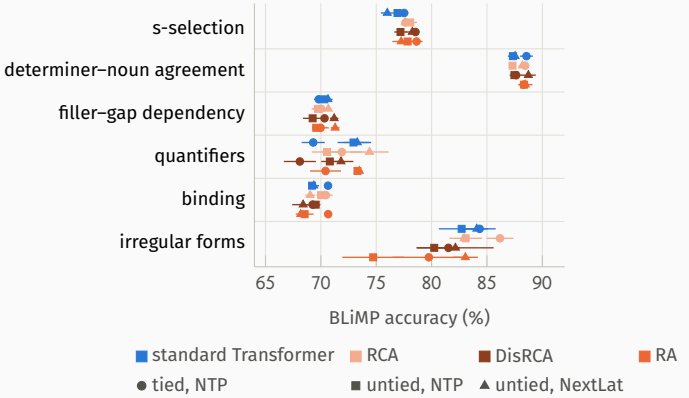
DAT gains most on islands, NPI licensing and subject–verb agreement



Rows ordered by mean DAT minus mean standard accuracy; 10M words, 512-token runs; mean over seeds, bars ±1 standard error. In each row, top to bottom: standard Transformer, RCA, DisRCA, RA.

- E.g. subject–verb agreement, the running example: standard Transformer 62.4–65.2%, DAT 67.5–70.0% over the three conditions.
- Architecture × linguistic domain: $\chi^2(12) = 402.68$, $p < 0.001$; DAT gains most on island effects, NPI licensing and subject–verb agreement (+0.25 to +0.29 log-odds), at most +0.11 elsewhere.

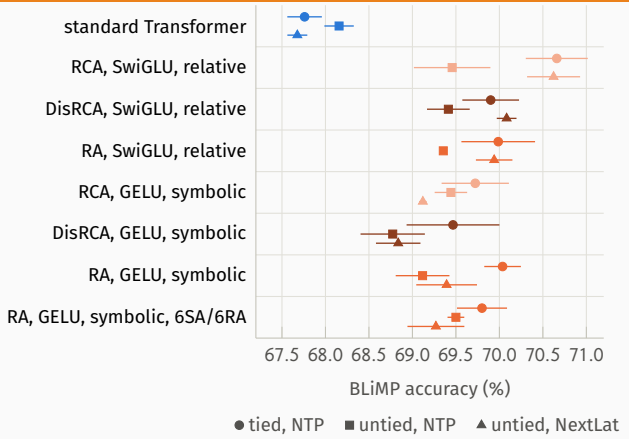
Smaller differences in the other six categories



Rows ordered by mean DAT minus mean standard accuracy; 10M words, 512-token runs; mean over seeds, bars ±1 standard error. In each row, top to bottom: standard Transformer, RCA, DisRCA, RA.

- Mean DAT minus mean standard Transformer accuracy: s-selection +1.08, determiner–noun agreement +0.29, filler–gap dependency −0.03, quantifiers −0.21, binding −0.40, irregular forms −2.20 points.
- The interaction model gives DAT at most +0.11 log-odds outside island effects, NPI licensing and subject–verb agreement.
- Irregular forms: the standard Transformer is above DisRCA and RA in each condition; RCA is above it in two of three.

BLiMP by configuration: all 21 DAT means are above the three standard means



10M words, 512-token runs; mean over 2 to 4 seeds, bars ±1 standard error. DAT rows: 9SA/3RA unless marked; relative: learned relative symbols; symbolic: symbolic attention.

- The models of the previous table: 12 layers, 768 dimensions, 12 heads; seven DAT configurations and the standard Transformer, each in three conditions.
- Lowest DAT mean: 68.77 (DisRCA, GELU, symbolic; untied head, NTP). Highest standard mean: 68.16 (untied head, NTP).
- Colours: the standard Transformer blue, the DAT types in orange shades, as in the BLiMP-categories slide; shapes: LM head and objective.

NextLat models fit human reading measures better

Main result

- Response: ΔR^2, the variance in reading times that model surprisal adds to the baseline predictors, per run and measure. Eleven measures: four eye-tracking, self-paced reading, six ERP components; only the first five enter the leaderboard Reading score.
- Objective: $t = 5.22$, $p < 0.001$. ΔR^2 is higher for NextLat than for NTP models.
- Model type: $t = -2.61$, $p = 0.010$. ΔR^2 is slightly higher for standard Transformers than for DAT models.
- The regression also controls for context length and for the reading condition (base or spillover regression).
- Our submitted NextLat entries also have higher Reading scores than our NTP entries: 7.12 vs. 6.56 at 10M words, 6.22 to 7.10 vs. 5.76 at 100M; these pairs also differ in head tying, or in size and curriculum.

Conclusion

Conclusion: architecture matters most for grammar, NextLat for reading

- Under the strict-small regime, architecture is the dominant factor for structural linguistic generalization: about 2.5 BLiMP points between the best DAT and the best standard Transformer (70.66% vs. 68.16%), at matched depth, width and head count.
- The gain comes from adding a relational stream rather than from enlarging it; at this scale the three relational attention types do not differ significantly on BLiMP.
- The auxiliary NextLat loss is secondary but significant and interacts with architecture: with NextLat, human reading measures are fit better (ΔR^2) and 5 of 7 finetuned (Super)GLUE accuracies are higher.
- Among symbol sources only the relational-symbolic variants underperform; the parameter-free RoPE-based relative symbols do not differ significantly from the learned library they replace.
- At 100M words full relational attention is higher on six of seven measures in a matched pair; our two strongest models rank 6th and 7th of 55 on the official strict leaderboard (16 October 2026).

Code, checkpoints and acknowledgments

- Code (model, training and evaluation scripts, configurations, analysis): `https://github.com/abrsvn/babylm_dat_2026`
- Submitted checkpoints: HuggingFace Hub, linked from the official leaderboard entries.

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